ACHELION CAPITAL MANAGEMENT, LLC

ARMS Addendum 4

Part One — The Governing Principle

Part Two — PTRH Coverage Adequacy Model (CAM)

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Read Part One before writing any code. The principle it contains is not context — it is the architectural directive that governs every decision you make about human touchpoints in ARMS.

"The PM needs to be at the lunch. The system needs to be running the portfolio."

PART ONE — THE GOVERNING PRINCIPLE

Why every human touchpoint you eliminate matters

0. In the PM's Own Words

> *The most important thing is keeping humans out of the process. Where the PM needs to be is not second-guessing or attempting to be smarter than the algorithm and AI — that is futile. The PM needs to be the boots on the ground, at the lunch, networking to find the next CEO that is selling or buying before it happens. The world has changed. The market is regressing to what used to add value: not illegal insider information but insider awareness that translates to moving before consensus by having a deep network of boots on the ground.*

That statement — made on March 27, 2026, the morning the Strait of Hormuz was formally closed and the ARMS regime score reached 0.79 — is the governing principle of every architectural decision in this system. It is not context. It is a directive.

Read it again before you build anything. Then ask: does this feature I am about to build put the PM closer to the lunch or further from it?

1. The Full Principle — Stated Precisely

1.1 What the PM's edge is not

The PM's edge is not analytical superiority over the algorithm. Every public signal — earnings calls, regulatory filings, price action, macro data, options flow, job postings, patent filings — is processed faster and more completely by ARMS than any human can read it. Attempting to out-analyze the system is not only futile — it is a misallocation of the most scarce resource in the enterprise: the PM's time and attention.

Every minute the PM spends confirming what the system already knows is a minute they are not building the network that creates the only durable edge left in modern markets.

1.2 What the PM's edge is

The PM's irreplaceable edge is physical presence in the rooms where the future is being decided before it enters any dataset.

- The lunch with the CEO who is quietly reducing their position — before the Form 4 is filed.
- The conversation with the TSMC supply chain executive whose capacity allocation decisions predict semiconductor demand two quarters ahead of any earnings call.
- The network that surfaces the insight before it becomes a signal — and the signal before it becomes a headline.
- Not illegal insider information. Earned, relationship-driven awareness that precedes consensus by weeks or months and cannot be replicated by any algorithm operating on public data.

This is Category A intelligence. It is what SENTINEL Gate 6 was built to capture. It is the only input the system cannot generate itself. Everything else ARMS can produce — and must produce — without human involvement.

1.3 The division of labor — stated as a law

The system executes everything that can be decided by math, process, and data inflow. The PM executes everything that requires being in the room where the data does not yet exist.

There are no exceptions to this division. A recurring decision queue item that requires the same PM action every day is not a safety feature. It is evidence of an incomplete build. A confirmation window for a deterministic outcome is not discipline. It is latency. A sizing decision made by PM opinion rather than MICS formula is not conviction. It is noise.

Every human touchpoint in ARMS must be tested against this question: does this require being in the room where the data does not yet exist? If the answer is no — automate it. If the answer is yes — preserve it, surface it precisely, and make the PM's response as frictionless as possible.

1.4 The ALAB lesson — why this principle was restated today

On March 21, 2026, a CEO insider sale was filed on a current portfolio position. The CDM and TDC modules that would have caught it on day one were not yet built. The signal appeared in the daily monitor as a manual flag. Six daily monitors carried the flag. The PM did not act. The position continued to deteriorate in a worsening market environment. The decision to trim — which the system would have initiated automatically within 24 hours of the Form 4 filing under the CDM/TDC specification — was deferred six days into worse conditions.

This is the cost of the gap. It is real, it is measurable, and it illustrates precisely why the governing principle exists. The PM should not have been watching the monitor waiting to make a sizing decision. The system should have made it. The PM should have been at the lunch — the one that might have surfaced the next CEO who is selling before the Form 4 is filed.

The CDM and TDC modules Josh builds will close this gap permanently. But the principle applies to every module, every decision, every human touchpoint across the entire ARMS architecture. Build everything with this case study in mind.

1.5 What Josh must ask before preserving any human touchpoint

If the answer is YES to this question	If the answer is NO to this question
Does this decision require knowledge that exists BEFORE the data — a personal conversation, a pattern recognition from lived experience, a pre-consensus insight no algorithm can generate?	Can this decision be made correctly using math, process, and data inflow that ARMS already has or can collect?
PRESERVE the human touchpoint. Surface it to the PM immediately. Make the response frictionless — one tap, structured question, specific context. Do not make the PM wait.	**AUTOMATE IT. Remove it from the decision queue. The PM should never see it. If the system fails to execute, fire an alert. The PM acts on system failures — not on system operations.**

> *The PM is in Dubai. The network is being built. The system is running the portfolio. This is not a future aspiration. This is the design target. Every line of code Josh writes should move toward it.*

PART TWO — PTRH COVERAGE ADEQUACY MODEL

CAM — Autonomous Tail Hedge Sizing · Addendum to Module A (PTRH Automation)

2. Why CAM Exists — The Gap It Closes

The PTRH sizing question — how much tail protection does the portfolio need right now — is a math question. It has always been a math question. ARMS has all the inputs. The PM should never be asked to answer it.

Module A (PTRH Automation, specified in Addendum 1) closes the execution gap: the system rolls puts automatically, resizes on regime transition, and corrects notional drift. What it does not do is autonomously determine whether the current coverage level is adequate given the specific risk configuration of the portfolio at any given moment.

The current specification requires a PM declaration of a dual-risk event before the system applies an override coverage standard. That declaration is a human touchpoint. It is asking the PM to answer a math question. The CAM removes that touchpoint entirely by calculating the required coverage from first principles using data ARMS already has.

File	src/engine/cam.py — new module, extends tail_hedge.py
Tier classification	Tier 0 — fully autonomous. No PM involvement. No declaration required. No veto window.
Build time	2-3 days — pure Python arithmetic on existing module outputs. No new APIs.
Dependencies	ARAS regime output · FEM factor exposure scores · Macro Compass composite inputs · Master Engine current weights · broker_api current QQQ put positions
Runs	Every 5 minutes with the data ingestion cycle — same cadence as ARAS
Ongoing cost	$0 — no new APIs or feeds
Replaces	The dual-risk PM declaration touchpoint in Module A. PM is never asked about PTRH sizing again under any conditions.

3. The Coverage Adequacy Formula

The CAM answers one question every 5 minutes: given the current portfolio composition, regime score, and identified risk factors, what QQQ put notional is required to protect the portfolio to the defined loss floor? It then compares that to actual notional and auto-corrects any shortfall.

3.1 Inputs — all available from existing ARMS modules

Input	Source module	Update frequency	What it represents
current_equity_pct	Master Engine	Every portfolio snapshot	Actual equity exposure as % of NAV at this moment
regime_score	ARAS	Every 5 min	Current composite risk score 0.0–1.0
fem_concentration_score	FEM	Daily + real-time on CDM trigger	Factor concentration level — how correlated the equity book is
macro_stress_score	Macro Compass	Every 5 min	Derived from Brent, 10Y, VIX, credit spreads — measures acute stress level 0.0–1.0
cdm_active_signals	CDM	Real-time	Number and severity of active cross-sector risk signals
nav	Master Engine	Every portfolio snapshot	Current portfolio NAV
current_put_notional	broker_api	Real-time	Actual QQQ put notional currently in place

3.2 The Coverage Requirement Formula

The required coverage is the product of four multipliers applied to a base coverage rate. Each multiplier reflects a specific risk dimension. When all four are elevated simultaneously — as they are today — the system automatically increases coverage without any human involvement.

```
def calculate_required_coverage(
    current_equity_pct:      float,  # e.g. 0.40 for 40% equity
    regime_score:            float,  # 0.0 - 1.0
    fem_concentration_score: float,  # 0.0 - 1.0 (ALERT = >0.80)
    macro_stress_score:      float,  # 0.0 - 1.0
    cdm_active_signals:      int,    # count of CRITICAL/HIGH signals
    nav:                     float   # current portfolio NAV
) -> float:
```

```
# Base coverage: 1.2% of NAV (minimum permanent insurance)
BASE_COVERAGE_PCT = 0.012

# Multiplier 1: Regime — higher score = more coverage needed
# Maps linearly from 1.0x at RISK_ON to 2.0x at CRASH boundary
regime_multiplier = 1.0 + (regime_score * 1.0)

# Multiplier 2: Equity exposure — more equity = more to protect
# Full deployment (100%) = 1.0x, current 40% ceiling = 0.67x floor
equity_multiplier = 0.67 + (current_equity_pct * 0.33)

# Multiplier 3: Concentration — correlated book = amplified loss
# NORMAL (<0.65): 1.0x · WATCH (0.65-0.80): 1.15x · ALERT (>0.80): 1.30x
if fem_concentration_score > 0.80:
    concentration_multiplier = 1.30
elif fem_concentration_score > 0.65:
    concentration_multiplier = 1.15
else:
    concentration_multiplier = 1.00

# Multiplier 4: Acute stress — macro + CDM signals
# Base macro stress component
stress_multiplier = 1.0 + (macro_stress_score * 0.40)
# Each active CDM CRITICAL/HIGH signal adds 3% to stress multiplier
stress_multiplier += (cdm_active_signals * 0.03)
stress_multiplier = min(stress_multiplier, 1.60)  # cap at 1.60x

# Required coverage percentage of NAV
required_pct = (BASE_COVERAGE_PCT
                * regime_multiplier
                * equity_multiplier
                * concentration_multiplier
                * stress_multiplier)

# Floor: never below regime-table minimum (from Module A)
regime_minimum = get_regime_table_minimum(regime_score)
required_pct = max(required_pct, regime_minimum)

# Ceiling: never above 3.5% of NAV (prevents over-hedging)
required_pct = min(required_pct, 0.035)

return required_pct * nav  # returns required notional in dollars
```

3.3 Worked example — current conditions (March 27, 2026)

This is the calculation ARMS would have run at 0530 CT this morning automatically.

Input	Current value	Multiplier result
Base coverage	1.2% of NAV	Starting point
Regime score (ARAS)	0.79	Regime multiplier: 1.0 + (0.79 × 1.0) = 1.79×
Equity exposure (Master Engine)	40% (DEFENSIVE ceiling)	Equity multiplier: 0.67 + (0.40 × 0.33) = 0.80×
FEM concentration (FEM module)	~0.82 (ALERT)	Concentration multiplier: 1.30×
Macro stress score (Macro Compass)	~0.85 (Brent $110, 10Y 4.45%, VIX 27)	Base stress: 1.34 + 2 CDM signals × 0.03 = 1.40×
Required coverage %	1.2% × 1.79 × 0.80 × 1.30 × 1.40	= 3.12% of NAV
vs. current 1.5× base	= 1.8% of NAV in puts	SHORTFALL: 1.32% of NAV — AUTO-CORRECT TRIGGERED

The CAM would have identified a 1.32% NAV shortfall in PTRH coverage this morning and auto-corrected to 3.12% before the market opened. The PM would never have been asked about it. The daily monitor would show: 'PTRH: CAM auto-corrected to 3.12% NAV at 0525 CT — regime 0.79, FEM ALERT, macro stress 0.85.' Status confirmation. Not a decision request.

3.4 Auto-correction logic

```
def run_cam_correction(required_notional: float,
                       current_notional: float,
                       nav:              float,
                       positions:        List[OptionsPosition]) -> CamResult:

    tolerance = 0.05  # 5% tolerance band — avoid constant micro-adjustments

    pct_diff = abs(required_notional - current_notional) / required_notional

    if pct_diff <= tolerance:
        return CamResult(action='NONE', reason='Within tolerance band')

    if required_notional > current_notional:
        # Increase coverage — buy additional puts at ATM
        delta = required_notional - current_notional
        execute_put_purchase(notional=delta, strike='ATM')
        log_cam_action('INCREASE', delta, required_notional, current_notional)
        return CamResult(action='INCREASED', delta=delta)
```

```
    else:
        # Decrease coverage — reduce puts (over-hedged)
        # Only reduce if over-hedged by more than 15% — asymmetric tolerance
        if pct_diff > 0.15:
            delta = current_notional - required_notional
            execute_put_sale(notional=delta)
            log_cam_action('DECREASED', delta, required_notional, current_notional)
            return CamResult(action='DECREASED', delta=delta)
        return CamResult(action='NONE', reason='Over-hedged within asymmetric tolerance')
```

Note the asymmetric tolerance: the system corrects under-coverage at 5% deviation but only corrects over-coverage at 15% deviation. This reflects the asymmetric cost of being wrong in each direction. Under-hedged in a crash is catastrophic. Over-hedged is merely slightly expensive.

4. What Disappears From the PM's Day

After CAM is built and deployed, the PM is never asked about PTRH sizing under any conditions — not for routine sizing, not for regime transitions, not for dual-risk scenarios, not for any market event.

Scenario	Before CAM	After CAM
Routine regime transition	PM confirms new sizing in decision queue	System auto-resizes. PM sees confirmation in module status.
Dual-risk event (Iran + Warsh)	PM declares event and sets coverage standard	CAM calculates required coverage from inputs. Auto-corrects. PM not involved.
Brent spikes overnight	Daily monitor flags PTRH sizing concern	CAM runs at 0525 CT. Auto-corrects before open. Monitor shows confirmation.
FEM reaches ALERT	PM asked to confirm PTRH adequacy	CAM reads FEM score, applies concentration multiplier, auto-corrects.
VIX spikes intraday	PM receives LAEP mode alert and PTRH note	CAM recalculates macro_stress_score on next 5-min cycle. Auto-corrects if needed.
Score approaches CRASH boundary	PM confirms PTRH adequacy in decision queue	CAM runs continuously. As score approaches 0.80, required coverage increases

		automatically. No PM notification unless execution fails.

5. Daily Monitor Integration

The PTRH module status card in the daily monitor is updated to show CAM output rather than a sizing confirmation request. The PM reads a status line, not an action item.

Monitor field — PTRH card	Shows: current notional % · required notional % (CAM) · last auto-correction timestamp · auto-correction reason · next scheduled review
Example — normal session	PTRH: 1.84% NAV · CAM required: 1.80% NAV · Within tolerance · Last correction: 0525 CT Mar 27 (regime 0.79, FEM ALERT)
Example — active correction	PTRH: AUTO-CORRECTED 0525 CT · 1.8% → 3.12% NAV · Reason: regime 0.79 + FEM ALERT + macro stress 0.85 · No action required
Alert condition	Only fires if execution FAILS — broker API error, insufficient liquidity, options market closed. PM acts on failures. Never on operations.
Decision queue	PTRH never appears in the PM decision queue after CAM deployment. Zero exceptions.

6. Updated File Structure

File	Status	Change
src/engine/cam.py	NEW	Coverage Adequacy Model — full autonomous PTRH sizing calculation and auto-correction. Runs every 5 minutes.
src/engine/tail_hedge.py	UPDATED	Imports CAM. Regime-table minimum becomes the floor in CAM formula. DTE roll and drift correction unchanged.
src/engine/macro_compass.py	UPDATED	Exposes macro_stress_score as a named output for CAM consumption. Composite of Brent

		deviation from $80 baseline, 10Y vs neutral rate, VIX level, credit spread.
src/reporting/daily_monitor.py	UPDATED	PTRH module status card shows CAM output fields. PTRH removed from decision queue template permanently.

7. Testing Protocol

1. Set regime_score to 0.79, fem_concentration_score to 0.82, macro_stress_score to 0.85, cdm_active_signals to 2. Verify CAM output matches the worked example in Section 3.3 — required notional should be approximately 3.12% of NAV.
2. Set all inputs to RISK_ON baseline (regime 0.25, FEM 0.40, macro stress 0.20, zero CDM signals). Verify required notional is close to base 1.2% NAV — the minimum permanent insurance floor.
3. Set required notional to 3.12% and current notional to 1.8%. Verify auto-correction fires and buys the additional puts. Verify session log captures the correction with all four multiplier values.
4. Set required notional to 1.8% and current notional to 1.9% (4.5% over-hedged — within asymmetric tolerance). Verify NO correction fires.
5. Set required notional to 1.8% and current notional to 2.2% (18% over-hedged — exceeds asymmetric tolerance). Verify reduction fires.
6. Simulate a broker API failure during correction. Verify DANGER alert fires in module status. Verify decision queue receives a single item: 'PTRH execution failed — manual review required.' This is the only PTRH item that should ever appear in the decision queue.

After CAM is live: the PM is never asked about tail hedge sizing again. Not once. The system knows what protection the portfolio needs. The system provides it. The PM is at the lunch.

"The PM needs to be at the lunch. The system needs to be running the portfolio."

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